

Computer Practical

Gaussian Plume Model

Atmospheric dispersion from point sources

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1 Overview

In this practical you will use a Gaussian plume model to investigate how pollutants emitted from an elevated point source are dispersed by the atmosphere. You will make a sequence of controlled comparisons involving wind direction, atmospheric stability, number of sources and aerosol water uptake.

The model is deliberately simple. It is designed to expose the physical assumptions rather than to reproduce a full regulatory air-quality model. Keep a practical record: save each figure using the suggested filename and note the settings used for the run.

An important modelling principle in this practical is to change one thing at a time. Where one result is reused as the reference case for the next comparison, you do not need to rerun it.

By the end of the practical you should be able to:

- explain the main assumptions behind a steady Gaussian plume model;
- relate wind direction and atmospheric stability to downwind pollutant concentrations;
- interpret the Pasquill–Gifford stability classes A–F and their effect on plume spread;
- explain how contributions from several point sources add together;
- distinguish dry particulate mass from an idealised humidified mass loading;
- recognise important limitations of a simple Gaussian plume calculation.

2 From advection–diffusion to a Gaussian plume

For a pollutant concentration C , the three-dimensional advection–diffusion equation can be written schematically as

$$\frac{\partial C}{\partial t} + u \frac{\partial C}{\partial x} + v \frac{\partial C}{\partial y} + w \frac{\partial C}{\partial z} = K_x \frac{\partial^2 C}{\partial x^2} + K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2}. \quad (1)$$

For the teaching model we assume a steady plume, a constant mean wind along the downwind direction, negligible mean crosswind and vertical advection, and negligible diffusion along the wind compared with advection. This leaves

$$u \frac{\partial C}{\partial x} = K_y \frac{\partial^2 C}{\partial y^2} + K_z \frac{\partial^2 C}{\partial z^2}. \quad (2)$$

For a continuous elevated point source of strength Q , one solution has the familiar Gaussian form

$$C(x, y, z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right]. \quad (3)$$

Here H is stack height and σ_y and σ_z describe the lateral and vertical spread of the plume. The second vertical exponential is an “image source” that represents reflection at the ground.

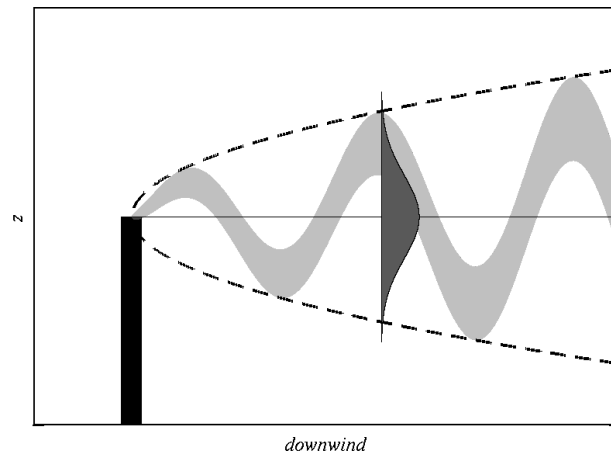


Figure 1: Schematic of the scenario being modelled, where H is the height of the stack.

Stability matters through σ_y and σ_z . The code uses empirical Pasquill–Gifford/ISC-style dispersion fits. Class A is very unstable and class F very stable. Unstable air generally produces faster plume spread; stable air suppresses vertical mixing and can keep a plume narrow for longer. Do not assume that “more unstable” automatically means “higher concentration” at every surface receptor: the answer depends on where the receptor lies relative to the source and plume.

3 Log in and download the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is 130.88.66.57; otherwise use the server details supplied for the course. Replace YOUR_USERNAME with your own username.

In your SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

Open a second terminal window for file transfer:

```
sftp YOUR_USERNAME@130.88.66.57
```

Your password will not be displayed while you type it.

File-transfer reminder. Keep the SSH window open for editing and running the model, and use the second SFTP window for retrieving your results. Commands such as `python3`, `nano` and `git` are entered in the **SSH window**; `get` is entered in the **SFTP window**.

In SFTP, the general form is:

```
get REMOTE_FILE LOCAL_FILENAME
```

The first filename is the file on the server. The second is the name given to the downloaded copy on your own computer.

Each new simulation overwrites the remote output image from the previous simulation. Download each figure immediately using the suggested local filename. The local files form part of your practical record.

In the SSH window, obtain the repository and enter the Python directory:

```
git clone https://github.com/EnvModelling/gaussian-plume-model-practical
cd gaussian-plume-model-practical/python
pwd
ls
```

You should see `gaussian_plume_model.py`, `gauss_func.py`, `calc_sigmas.py`, `overlay_on_map.py` and `map_green_lane.mat`.

4 Inspecting and running the model

Open the main script with line numbers displayed:

```
nano -l gaussian_plume_model.py
```

In nano, `Ctrl-W` searches. Search for `PARAMETERS TO CHANGE`. In the current repository the editable block is around lines 79–101. Line numbers are approximate; searching for a parameter name is more reliable if the code is updated.

Run the model with:

```
python3 gaussian_plume_model.py
```

The model prints the full path of the output it creates. The default run writes

```
/tmp/YOUR_USERNAME/plan_view.png
```

In the SFTP window, download it and give the local copy a useful name:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_default.png
```

Here the server file remains `plan_view.png`, while the copy on your own computer is saved as `gpm_default.png`.

Controlled comparisons. The parameter `RANDOM_SEED = 22001` makes the synthetic wind sequence reproducible. Keep this seed unchanged throughout the comparisons so that you do not change the meteorology at the same time as the parameter you are trying to study.

At the end of the practical you can restore the edited script with:

```
git restore gaussian_plume_model.py
```

This restores that tracked file without affecting figures already saved in `/tmp/$USER/` or downloaded to your computer.

5 Parameters used in the practical

Table 1: Main editable parameters in `gaussian_plume_model.py`.

Parameter	Default	Meaning
RH	0.90	Relative humidity used by the idealised water-uptake calculation
AEROSOL_TYPE	SODIUM_CHLORIDE	Particle composition
HUMIDIFICATION	DRY_AEROSOL	Use dry mass or apply idealised water uptake
STABILITY_CLASS	1	Pasquill–Gifford class: 1=A through 6=F
STABILITY_USED	CONSTANT_STABILITY	Fixed class or idealised annual cycle
OUTPUT	PLAN_VIEW	Plan view, vertical slice, receptor time series or map overlay
X_SLICE	25	Python grid index 25 corresponds to $x = 0$ m
Y_SLICE	0	Python grid index 0 corresponds to $y = -2500$ m
WIND	PREVAILING_WIND	Constant, random or prevailing wind direction
WIND_SPEED	5.0 m s^{-1}	Wind speed; held fixed in these experiments
RANDOM_SEED	22001	Seed for reproducible synthetic wind directions
STACKS	ONE_STACK	Number of active stacks (1–3)
Q	40 g s^{-1} each	Dry pollutant mass emission rate
H	50 m each	Stack heights
DAYS	50	Number of 24-hour periods included in the run

Table 2: Pasquill–Gifford classes used by the teaching model.

Value	Class	Description
1	A	very unstable
2	B	moderately unstable
3	C	slightly unstable
4	D	neutral
5	E	moderately stable
6	F	very stable

Table 3: Output modes and remote files.

Setting	What is plotted	Remote output
PLAN_VIEW	time-mean surface concentration	plan_view.png
HEIGHT_SLICE	time-mean y–z concentration slice	height_slice.png
SURFACE_TIME	concentration at one surface receptor	surface_time.png
OVERLAY	mean surface contours on the supplied map	overlay.png

6 Experiments

For every experiment, record the settings you changed, save the output using the suggested local filename, and write one or two sentences describing what changed physically. Unless instructed otherwise, leave all unlisted parameters unchanged.

6.1 Wind direction and variability

First isolate the effect of wind direction. Set:

```
STABILITY_CLASS = 4
STABILITY_USED = CONSTANT_STABILITY
STACKS = ONE_STACK
OUTPUT = PLAN_VIEW
HUMIDIFICATION = DRY_AEROSOL
DAYS = 50
```

Keep RANDOM_SEED fixed.

Run the following two cases, changing only WIND:

Run	Set WIND to	Suggested local filename
1	CONSTANT_WIND	gpm_1stack_constant.png
2	PREVAILING_WIND	gpm_1stack_prevailing.png

For example, after the constant-wind run use:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_1stack_constant.png
```

and after the prevailing-wind run use:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_1stack_prevailing.png
```

Prediction. Before comparing the plots, predict the shape expected for each wind assumption.

Exercise. Explain how wind-direction variability changes the area exposed to pollutant and the maximum time-mean concentration. Why does the prevailing-wind case differ from a constant-wind plume?

Optional extension. If time permits, set WIND = FLUCTUATING_WIND, save the result as gpm_1stack_random.png, and compare a completely random wind direction with the prevailing-wind case.

6.2 Multiple stacks

Your previous prevailing-wind calculation already provides the **one-stack reference case**: `gpm_1stack_prevaling.png`. Do not rerun it.

Keep all other settings unchanged and change only:

```
STACKS = THREE_STACKS
```

Run the model and download:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_3stacks_prevaling.png
```

The source positions, emissions and heights are stored in `STACK_X`, `STACK_Y`, `Q` and `H`. The first `STACKS` entries are used.

Exercise. Compare `gpm_1stack_prevaling.png` with `gpm_3stacks_prevaling.png`. Where do the concentration fields from the separate stacks overlap? In the code, concentrations from separate stacks are simply added. Explain why this superposition is possible for this linear plume model.

Optional extension. If time permits, try `TWO_STACKS`, move one stack, or change one stack height. Predict the effect before rerunning.

6.3 Atmospheric stability: surface plan view

The one-stack prevailing-wind result `gpm_1stack_prevaling.png` is also your **neutral (class D) reference case**. There is no need to rerun class D.

Set the following baseline:

```
OUTPUT = PLAN_VIEW
STACKS = ONE_STACK
WIND = PREVAILING_WIND
STABILITY_USED = CONSTANT_STABILITY
HUMIDIFICATION = DRY_AEROSOL
DAYS = 50
```

Now run only the two end-member stability cases:

Class	STABILITY_CLASS	Interpretation	Suggested local filename
A	1	very unstable	<code>gpm_stability_A_plan.png</code>
D	4	neutral	use <code>gpm_1stack_prevaling.png</code>
F	6	very stable	<code>gpm_stability_F_plan.png</code>

For class A:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_stability_A_plan.png
```

For class F:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_stability_F_plan.png
```

Exercise. Compare classes A, D and F. How do plume width, downwind extent and ground-level concentration change with atmospheric stability? Link your explanation to the behaviour of σ_y and σ_z . Is “more unstable” always the same as “higher concentration at every ground receptor”? Explain why not.

6.4 Atmospheric stability: vertical structure

Now compare the vertical structure of the two end-member cases. Set:

```

OUTPUT = HEIGHT_SLICE
X_SLICE = 25
STACKS = ONE_STACK
WIND = PREVAILING_WIND
STABILITY_USED = CONSTANT_STABILITY

```

so that the model plots a y - z slice at $x = 0$ m.

Run class A and class F:

Class	STABILITY_CLASS	Suggested local filename
A	1	gpm_stability_A_vertical.png
F	6	gpm_stability_F_vertical.png

Download each `height_slice.png` immediately after it is produced, for example:

```
get /tmp/YOUR_USERNAME/height_slice.png gpm_stability_A_vertical.png
```

Exercise. How does stability alter the vertical spread of the plume and the fraction of pollutant that reaches the surface? Relate the results qualitatively to the textbook plume regimes often called looping, coning and fanning. Note that this is a steady, time-mean Gaussian model: it changes plume spread but does not resolve individual turbulent “loops” in the instantaneous plume.

6.5 An idealised annual cycle in stability

The model can impose a deliberately simple seasonal cycle: stable classes in winter and unstable classes in summer. It does *not* include a diurnal stability cycle.

Set:

```

OUTPUT = SURFACE_TIME
STABILITY_USED = ANNUAL_CYCLE
STACKS = ONE_STACK
WIND = PREVAILING_WIND
X_SLICE = 25
Y_SLICE = 0
DAYS = 365

```

This receptor is at $x = 0$ m, $y = -2500$ m. Run the model and download:

```
get /tmp/YOUR_USERNAME/surface_time.png gpm_annual_cycle.png
```

The upper panel shows hourly concentration plus a 24-hour moving mean; the lower panel shows the imposed stability class.

Exercise. Describe the seasonal pattern at this receptor. Use both distance from the stack and plume spread to explain why the relationship between atmospheric instability and concentration at a particular point is not necessarily monotonic. What is realistic about this experiment, and what is intentionally artificial?

6.6 Idealised aerosol water uptake and map overlay

The original teaching code included a simplified humidity calculation. It is retained here, but it should not be described as a full Köhler model: the Kelvin curvature term and non-ideal solution effects are omitted. Under the ideal-solution approximation used here, the wet/dry particle-mass factor is

$$G_m = \frac{m_{\text{wet}}}{m_{\text{dry}}} = 1 + \frac{RH}{1 - RH} \nu \frac{M_w}{M_s}, \quad (4)$$

where ν is an idealised van't Hoff factor, M_w is the molar mass of water and M_s is the molar mass of the dry solute. The factor multiplies the plume mass concentration; it does not change the dispersion itself.

Set the following baseline:

```

OUTPUT = OVERLAY
STABILITY_CLASS = 4
STABILITY_USED = CONSTANT_STABILITY
STACKS = ONE_STACK
WIND = PREVAILING_WIND
AEROSOL_TYPE = SODIUM_CHLORIDE
RH = 0.90
DAYS = 50

```

Run only the dry and humidified sodium-chloride cases:

Case	HUMIDIFICATION	Suggested local filename
Dry	DRY_AEROSOL	gpm_nacl_dry.png
Humidified	HUMIDIFY	gpm_nacl_wet.png

Download the two overlay.png files as:

```
get /tmp/YOUR_USERNAME/overlay.png gpm_nacl_dry.png
```

and

```
get /tmp/YOUR_USERNAME/overlay.png gpm_nacl_wet.png
```

When humidification is enabled, the script prints the wet/dry mass factor used.

Prediction. Using $RH = 0.90$, calculate G_m for sodium chloride using $\nu = 2$ and $M_s = 58.44 \text{ g mol}^{-1}$. Predict whether humidification should change the shape of the plume contours, their magnitude, or both.

Exercise. Compare the dry and humidified maps. Does water uptake change the dispersion pattern itself? Explain the result using the way G_m is applied in the model.

Optional extension. Set $AEROSOL_TYPE = ORGANIC_ACID$, with $HUMIDIFICATION = HUMIDIFY$, and save the output as `gpm_organic_wet.png`. Before running, calculate the expected wet/dry mass factor using $\nu = 1$ and $M_s = 200 \text{ g mol}^{-1}$ and compare it with sodium chloride.

7 What this model leaves out

A useful model is defined as much by its assumptions as by its equations. This practical model does **not** include plume rise from stack temperature or momentum, terrain, buildings and wake effects, deposition, chemical reactions, changing wind speed with height, measured meteorology, or a realistic boundary layer. The source is continuous and the Gaussian equation represents a steady mean plume for each hourly meteorological state.

What you should take away. A Gaussian plume is a compact way to connect emissions, wind and turbulent dispersion to concentrations. The calculation can be very informative, but its predictions are conditional on the assumptions used for u , σ_y , σ_z , source geometry and meteorology. When you interpret a model result, always ask which process was represented, which was parameterised, and which was omitted.